

K-THEORY OF THE AFFINE BOUNDARY QUOTIENT: THE RETURN OF THE NORM AND THE TRACE

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ABSTRACT. For the affine control system of Paper II (its §6) at a *principal* prime $\mathfrak{p} = (\pi)$ of a number field K of degree n , with $q = N\mathfrak{p}$, we compute the K-theory of the boundary quotient $\partial\mathcal{A}^{\text{aff}}$. The tight quotient is stably isomorphic to $C_0(K_{\mathfrak{p}}) \rtimes (\mathcal{O}_K[1/\pi] \rtimes_{\pi} \mathbb{Z})$, and an iterated Pimsner–Voiculescu computation over the exterior-power tower $M_r = \varinjlim (\Lambda^r \mathcal{O}_K, t_r)$, with integral transfer $t_r = \pm(\Lambda^{n-r}\pi)^{\vee}$ and scaling action $\alpha_* = q^{-1}\Lambda^r\pi$, yields: a copy of $\mathbb{Z}$ in each of K_0 and K_1 from the top layer when $N_{K/\mathbb{Q}}\pi = +q$ (a single $\mathbb{Z}$ and a $\mathbb{Z}/2$ when $N\pi = -q$); the class of the unit generating a torsion summand $\mathbb{Z}/(q-1)$ of exact order $q-1$ — the norm survives, and the divisibility bound $(N\mathfrak{p}-1)[1] = 0$ of Paper II is attained; and, in the intermediate layers $0 < r < n$, finite torsion of order the prime-to- q part of $|\det(q \cdot 1 - \Lambda^r \pi)|$ up to primes inverted by the tower — for K real quadratic, a summand $\mathbb{Z}/|q \pm 1 - \text{Tr } \pi|$ of K_1 . The trace of the uniformizer enters the K-theory. Consequently the affine boundary is *not* a canonical invariant of $\mathfrak{p}$: replacing π by $u\pi$ for a unit u changes the torsion, computed explicitly in $\mathbb{Q}(\sqrt{2})$. The class group is located precisely at the entrance: for non-principal $\mathfrak{p}$ the semigroup of Paper II's §6 is not well defined over $\mathcal{O}_K$, and the ray-class repair is the first of three problems recorded honestly at the end.

1. THE SETTING, AND A REPAIR

Paper II (its §6) attaches to a set S of primes of K the affine system $\mathcal{A}_{K,S}^{\text{aff}} = C(\widehat{\mathcal{O}}_S) \rtimes (\mathcal{O}_K \rtimes J_S)$, translations by $\mathcal{O}_K$ and scalings by fixed uniformizers, and its boundary quotient $\partial\mathcal{A}_{K,S}^{\text{aff}}$, obtained by imposing, for each $\mathfrak{p} \in S$, the Cuntz relation $\sum_{a \in \mathcal{O}_K/\mathfrak{p}} \mu_{(a,\mathfrak{p})} \mu_{(a,\mathfrak{p})}^* = 1$; the quotient is a unital Kirchberg algebra satisfying the UCT, and $(N\mathfrak{p}-1)[1] = 0$ in K_0 .

One repair is needed before any computation. If the uniformizer $\pi_{\mathfrak{p}}$ is merely local, the semigroup law $(b, \mathfrak{a})(b', \mathfrak{a}') = (b + \pi_{\mathfrak{a}}b', \mathfrak{a}\mathfrak{a}')$ leaves $\mathcal{O}_K$: the affine monoid of Paper II's §6 is well defined over $\mathcal{O}_K$ only when the scalings are by *global* elements. We therefore assume throughout that $\mathfrak{p} = (\pi)$ is principal with a chosen generator $\pi \in \mathcal{O}_K$, and we work at a single prime, $S = \{\mathfrak{p}\}$, $q := N\mathfrak{p}$, $n := [K : \mathbb{Q}]$, $N_{K/\mathbb{Q}}\pi = \pm q$. The non-principal case requires a ray-class extension of the acting monoid and is recorded as Problem (a) in §4; as Remark 3.5 shows, this is exactly where the class group enters the affine theory.

2. THE DILATED FORM

Proposition 2.1. $\partial\mathcal{A}_{K,\{\mathfrak{p}\}}^{\text{aff}} \otimes \mathbb{K} \cong C_0(K_{\mathfrak{p}}) \rtimes (\mathcal{O}_K[1/\pi] \rtimes_{\pi} \mathbb{Z})$, where $\mathcal{O}_K[1/\pi]$ acts by translation through its dense image (strong approximation) and $\mathbb{Z}$ by multiplication by π . In particular the two algebras have the same K-theory.

Proof sketch. On the boundary the scaling isometry μ_{π} has the Cuntz relation above, and the standard dilation of the corner endomorphism $\text{Ad } \mu_{\pi}$ to an automorphism enlarges the base $\mathcal{O}_{\mathfrak{p}}$ to $K_{\mathfrak{p}} = \bigcup_m \pi^{-m} \mathcal{O}_{\mathfrak{p}}$ and the translation group $\mathcal{O}_K$ to $\mathcal{O}_K[1/\pi]$; the image of $\mathcal{O}_K$ is dense in $\mathcal{O}_{\mathfrak{p}}$

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Paper XXX of the series on localized Bost–Connes systems. The framework is that of [1], cited by DOI; Paper II of the series, referred to by numeral, is a published companion (its §6 introduces the affine system), and the facts used from it are restated in place.

(strong approximation), hence $\mathcal{O}_K[1/\pi]$ is dense in $K_{\mathfrak{p}}$, the action is minimal, and $1_{\mathcal{O}_{\mathfrak{p}}}(C_0(K_{\mathfrak{p}}) \rtimes (\mathcal{O}_K[1/\pi] \rtimes \mathbb{Z}))1_{\mathcal{O}_{\mathfrak{p}}}$ is a full corner isomorphic to $\partial\mathcal{A}_{K,\{\mathfrak{p}\}}^{\text{aff}}$. The group $\mathcal{O}_K[1/\pi] \rtimes \mathbb{Z}$ is solvable, hence amenable, so the crossed product is nuclear and the Baum–Connes machinery below is available. $\square$

3. THE EXTERIOR TOWER AND THE ITERATED PIMSNER–VOICULESCU COMPUTATION

Write $B = C_0(K_{\mathfrak{p}}) \rtimes \mathcal{O}_K[1/\pi]$ and α for the scaling automorphism, so that the algebra of Proposition 2.1 is $B \rtimes_{\alpha} \mathbb{Z}$.

Lemma 3.1 (The tower). *For $0 \leq r \leq n$ let $t_r : \Lambda^r \mathcal{O}_K \rightarrow \Lambda^r \mathcal{O}_K$ be the transfer, characterized by $t_r \circ \Lambda^r(\pi) = q \cdot \text{id}$; it is integral, $t_r = \pm(\Lambda^{n-r}\pi)^{\vee}$ under the duality $\Lambda^r \otimes \Lambda^{n-r} \rightarrow \Lambda^n \cong \mathbb{Z}$ (multiplication by π on Λ^n being $N\pi$). Then*

$$K_*(B) \cong \bigoplus_{r \equiv * (2)} M_r, \quad M_r := \varinjlim (\Lambda^r \mathcal{O}_K \xrightarrow{t_r} \Lambda^r \mathcal{O}_K \xrightarrow{t_r} \cdots),$$

and the scaling acts on M_r by the invertible map $\alpha_* = q^{-1}\Lambda^r\pi$ (inverse $q^{-1}t_r$).

Proof sketch. $C(\mathcal{O}_{\mathfrak{p}}) = \varinjlim_m C(\mathcal{O}_K/\mathfrak{p}^m)$ equivariantly; the translation action of $\mathcal{O}_K$ on $\mathcal{O}_K/\mathfrak{p}^m$ is transitive with stabilizer $\mathfrak{p}^m = \pi^m \mathcal{O}_K$, so by Green’s imprimitivity $C(\mathcal{O}_K/\mathfrak{p}^m) \rtimes \mathcal{O}_K \cong M_{q^m}(C^*(\pi^m \mathcal{O}_K))$, whose K-theory is the exterior algebra $\Lambda^*(\pi^m \mathcal{O}_K)$ in the usual parity. The connecting map from level m to level $m+1$ is the transfer for the index- q inclusion $\pi^{m+1} \mathcal{O}_K \subset \pi^m \mathcal{O}_K$, which under the identifications $\Lambda^r(\pi^m \mathcal{O}_K) \cong \Lambda^r \mathcal{O}_K$ (via $\Lambda^r \pi^{-m}$) becomes t_r ; passing to B replaces the colimit over m by the same colimit, the corner being full. Integrality of t_r : pairing with Λ^{n-r} and using $\Lambda^r \pi \otimes \Lambda^{n-r} \pi \mapsto N\pi = \pm q$ on Λ^n gives $t_r = \pm(\Lambda^{n-r}\pi)^{\vee}$. The scaling conjugates level m to level $m+1$ composed with $\Lambda^r \pi$, which in the colimit is $q^{-1}\Lambda^r \pi$; the two calibrations of Remark 3.2 pin the normalization. $\square$

Remark 3.2 (Two calibrations, $n = 1$). For $K = \mathbb{Q}$, $\pi = p = q$: at $r = 0$, $t_0 = q$ and $M_0 = \mathbb{Z}[1/q]$ with $\alpha_* = 1/q$ — the class $[1_{\mathcal{O}_{\mathfrak{p}}}]$ scales by the measure factor, as it must. At $r = 1$, $t_1 = q \cdot p^{-1} = 1$ and $M_1 = \mathbb{Z}$ with $\alpha_* = p/q = 1$: the unitary of a single translation coset has, in the merged level, class 1 and *not* p — this is the pivot of the whole computation, checked level by level on the Bunce–Deddens tower: the level- m generator of $K_1(M_{p^m}(C^*(p^m \mathbb{Z}))) \cong \mathbb{Z}$ is the coset unitary, the transfer to level $m+1$ carries it to the coset unitary again, and the scaling permutes the p cosets without multiplying the class. Anchors: the resulting $K_*(B \rtimes \mathbb{Z})$ below reproduces, for $K = \mathbb{Q}$, $K_0 = \mathbb{Z} \oplus \mathbb{Z}/(p-1)$ with $[1]$ generating the torsion and $K_1 = \mathbb{Z}$; at $p = 2$ this gives $K_0 \cong \mathbb{Z}$, $K_1 \cong \mathbb{Z}$, $[1] = 0$, consistent with the 2-adic ring algebra computation of Larsen–Li [3].

Theorem 3.3 (K-theory of the affine boundary, principal single prime). *Assume $\pi^A \neq q$ for every r -subset A of the archimedean embeddings and every $0 < r < n$, where $\pi^A = \prod_{i \in A} \pi^{(i)}$. Then, with $T_r := \text{coker}(1 - \alpha_* | M_r)$ finite for $0 < r < n$ and $T_0 = \mathbb{Z}/(q-1)$:*

$$N\pi = +q : \quad K_0(\partial\mathcal{A}^{\text{aff}}) \cong \mathbb{Z} \oplus \bigoplus_{r \text{ even}} T_r, \quad K_1(\partial\mathcal{A}^{\text{aff}}) \cong \mathbb{Z} \oplus \bigoplus_{r \text{ odd}} T_r,$$

$$N\pi = -q : \quad \text{the two } \mathbb{Z}\text{'s are replaced by a single } \mathbb{Z}/2 \text{ in parity } n.$$

The class $[1]$ generates $T_0 = \mathbb{Z}/(q-1)$: its order is exactly $q-1$, attaining the divisibility bound of Paper II. The order of T_r is the part of $|\det(q \cdot 1 - \Lambda^r \pi)|/q^{\binom{n}{r}}$ prime to the primes inverted by the tower t_r .

Proof sketch. Pimsner–Voiculescu [4] for $B \rtimes_{\alpha} \mathbb{Z}$: $0 \rightarrow \text{coker}(1 - \alpha_* | K_*(B)) \rightarrow K_*(B \rtimes \mathbb{Z}) \rightarrow \ker(1 - \alpha_* | K_{*+1}(B)) \rightarrow 0$, summed over the parity decomposition of Lemma 3.1. At $r = 0$: $1 - \alpha_* = 1 - q^{-1}$ on $\mathbb{Z}[1/q]$ has cokernel $\mathbb{Z}/(q-1)$ generated by the class of $1_{\mathcal{O}_{\mathfrak{p}}}$, and kernel 0.

At $r = n$: $M_n = \mathbb{Z}$, $\alpha_* = N\pi/q = \pm 1$; for $+1$ both kernel and cokernel are $\mathbb{Z}$, contributing the two free summands; for -1 , $1 - \alpha_* = 2$, cokernel $\mathbb{Z}/2$, kernel 0. At $0 < r < n$: the eigenvalues of α_* are π^A/q , none equal to 1 by hypothesis, so $1 - \alpha_*$ is injective with finite cokernel of the stated order; the extensions split at the level of orders and free ranks, the remaining extension data being Problem (b) of §4. $\square$

Example 3.4 (Real quadratic, explicitly). $n = 2$, π' the conjugate, $t_1 = \pm\pi'$ (multiplication by the conjugate — the transfer *is* the adjugate), $M_1 = \mathcal{O}_K$ with the primes dividing $N\pi'$ inverted, and $\alpha_* = \pi/q$. Then

$$\det(1 - \alpha_*) = N_{K/\mathbb{Q}}\left(1 - \frac{\pi}{q}\right) = \frac{N(q - \pi)}{q^2} = \frac{q \mp 1 - \text{Tr } \pi}{q} \cdot (\pm 1),$$

so $K_1(\partial\mathcal{A}^{\text{aff}})$ contains the summand $\mathbb{Z}/|q \pm 1 - \text{Tr } \pi|'$ (prime-to- q part), the sign that of $N\pi = \pm q$. *The trace of the uniformizer enters the K-theory.* Concretely, in $K = \mathbb{Q}(\sqrt{2})$ with $\mathfrak{p} = (3 + \sqrt{2})$, $q = 7$, $N\pi = +7$, $\text{Tr } \pi = 6$: the summand is $\mathbb{Z}/|7 + 1 - 6| = \mathbb{Z}/2$.

Remark 3.5 (The units are audible; the class group is at the door). Replacing π by $u\pi$ for $u \in \mathcal{O}_K^\times$ of norm 1 preserves $\mathfrak{p}$ and q but changes Tr : in $\mathbb{Q}(\sqrt{2})$, $u = 3 + 2\sqrt{2}$ gives $u\pi = 13 + 9\sqrt{2}$, $N = 7$, $\text{Tr} = 26$, and the summand becomes $\mathbb{Z}/|7 + 1 - 26| = \mathbb{Z}/18$. The affine boundary quotient is therefore *not* a canonical invariant of the prime: it hears the unit orbit of the chosen uniformizer — the reason Paper II's §6 had to “fix a family of uniformizers”. And the class group is located exactly at the entrance of the theory rather than inside the K-groups: it obstructs the existence of a global π at all, Problem (a) below.

4. WHAT THE AFFINE BOUNDARY REMEMBERS, AND WHAT REMAINS

Three layers of arithmetic survive where the multiplicative system forgets: the *norm*, as the exact order $q - 1$ of $[1]$; the *splitting data* of $\mathfrak{p}$, through the values of the exterior characteristic polynomials of π at q (for real quadratic fields, $\text{Tr } \pi$); and, unexpectedly, the *unit orbit* of the uniformizer, which makes the invariant finer than the prime itself. The degree n sets the number of layers; the unit *rank* enters only through the orbit sensitivity.

Three problems remain and are recorded as such. (a) *The ray-class repair*: for non-principal $\mathfrak{p}$ the acting monoid must be enlarged (ray classes, or a principalizing extension), and the resulting boundary compared with the principal case — this is where the class group genuinely enters. (b) *Extension data*: for $n \geq 3$, and for $|S| \geq 2$ under the iterated Pimsner–Voiculescu sequences, the orders and free ranks above determine K_* only up to group extensions. (c) *The q -part*: the torsion orders are computed prime to the primes inverted by the transfers; the exact q -primary bookkeeping requires the integral colimit, not just its localization.

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